Cantera vs NN | Test IC 44 | NRMSE [%]

T0 = 1031.1 K P0 = 2.272 atm phi0 = 0.770 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	36.2	24.8	9.96	14.2	25.2	370	34	9.57	8.29	N/A	Complete
2 P-only DeepONet10	**10k**	39.5	25.3	37.6	14.1	27.7	334	31.4	15.7	9.28	N/A	Complete
2 P-only DeepONet10	**long**	29.1	29.3	3.87	5.66	29.3	20.8	29.2	8.93	6.94	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	30.3	27.8	7.8	14.5	32.9	343	29.7	8.93	6.63	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	27.2	27.5	4.48	5.37	27.4	19.7	27.4	9.22	7.29	N/A	Complete
4 T/P DeepONet10	**10k**	26.5	28.2	19	6.91	28.6	43.2	27	93.4	5.51e+03	N/A	Complete
4 T/P DeepONet10	**long**	59	32.4	76.2	7.03	41.9	568	39.3	9.35	6.81	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	122	35	255	13.7	52.5	41.1	68.7	1.91e+03	2.52e+04	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	59	31.4	66.2	10.2	43	633	40.5	9.5	6.92	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	93.7	93.7	4.41	14.2	93.7	64.4	93.7	8.64	6.73	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	26.2	26.3	3.03	6.33	26.4	17.8	26.3	9.51	8.03	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	93.7	93.7	4.41	14.2	93.7	64.4	93.7	8.63	6.73	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	23.4	23.6	3.21	5.99	23.7	20	23.5	9.31	8.32	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.